I Used to Hate Go

What changed my mind as a Java/Kotlin developer

About me!

Kubilay Karpat

- Software Consultant at Xebia
- Backend development
- Java, Kotlin and Go

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1 Where I Start

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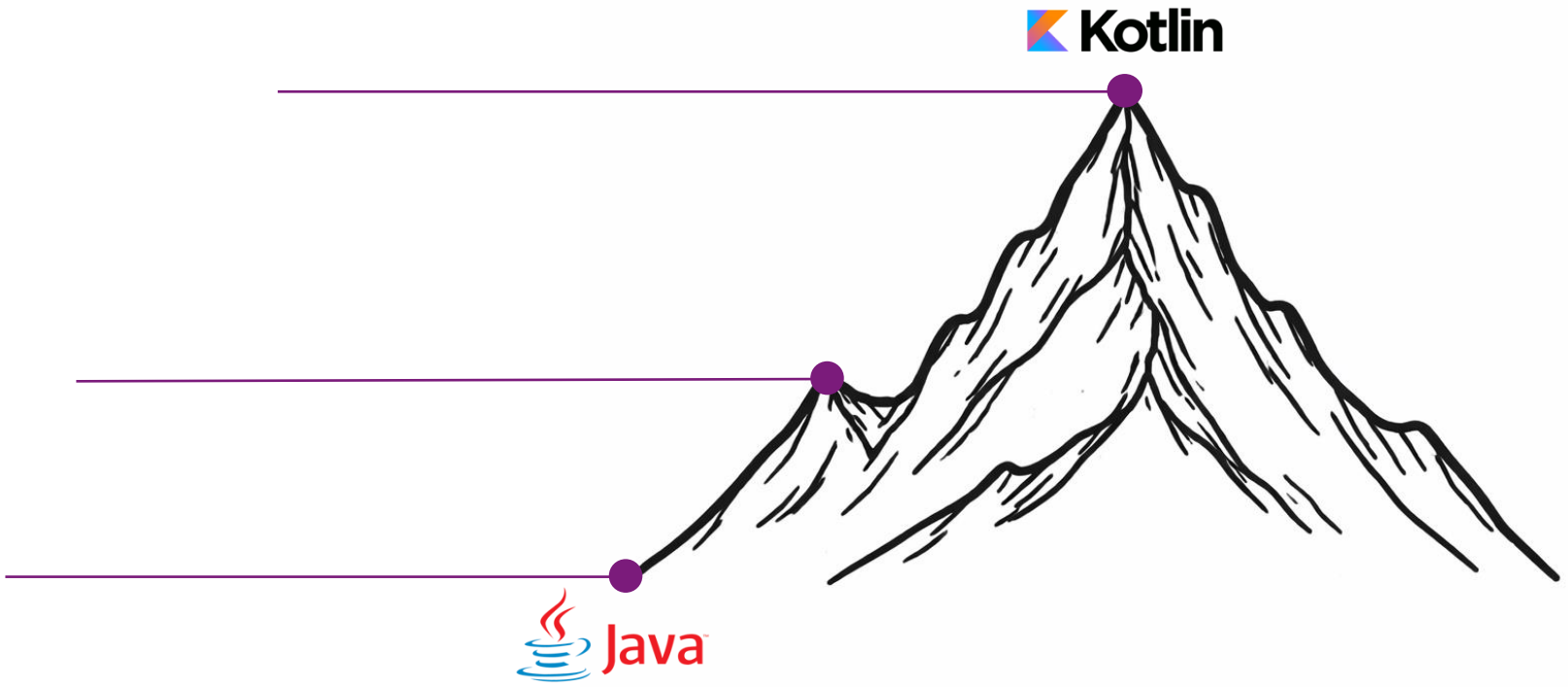
My Journey to the “peak”

2021: Backend Kotlin developer
Cashback/savings apps

2020: Backend Java developer
Payment provider

2016: Full stack Java developer
Jira plugin development



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Why I didn’t like Go?

- Verbose
- Fewer abstractions
- Smaller ecosystem

It felt like ”going backwards”

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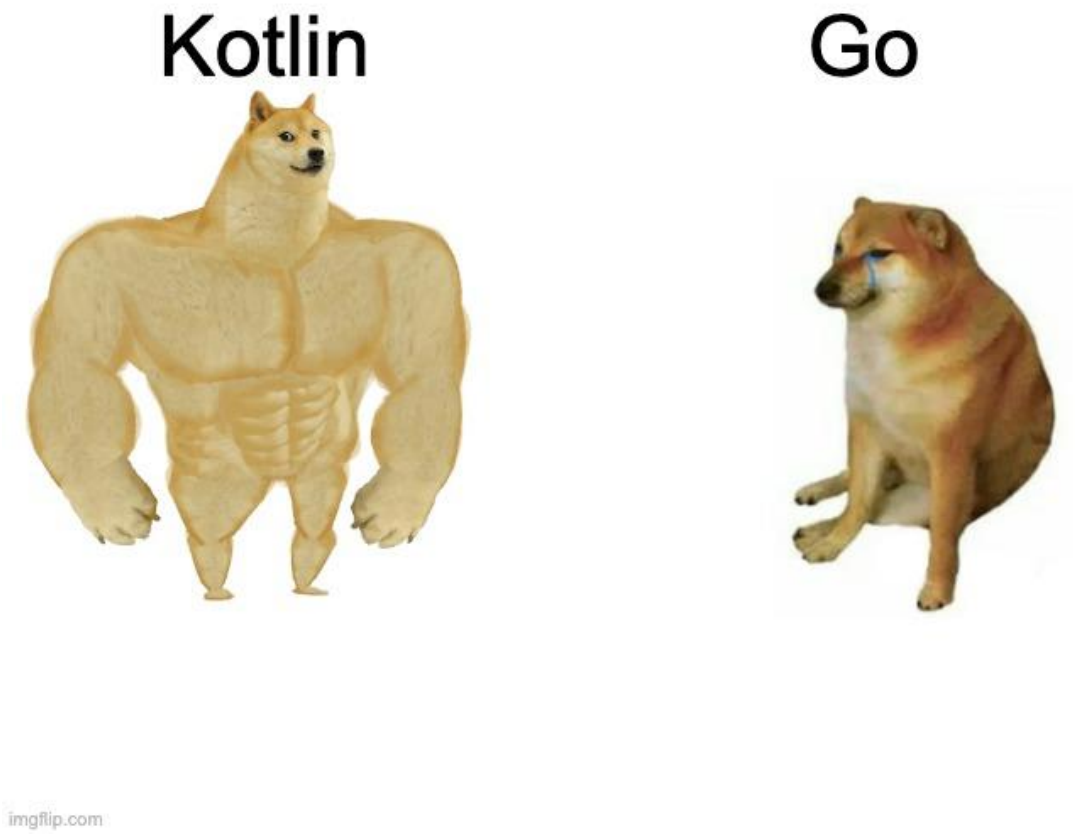
How it felt back then



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Facing my enemies

- In 2024 I started working at Xebia as a software consultant
- Only available project was a Go project
- I wasn’t sure how long I have to wait for another opportunity

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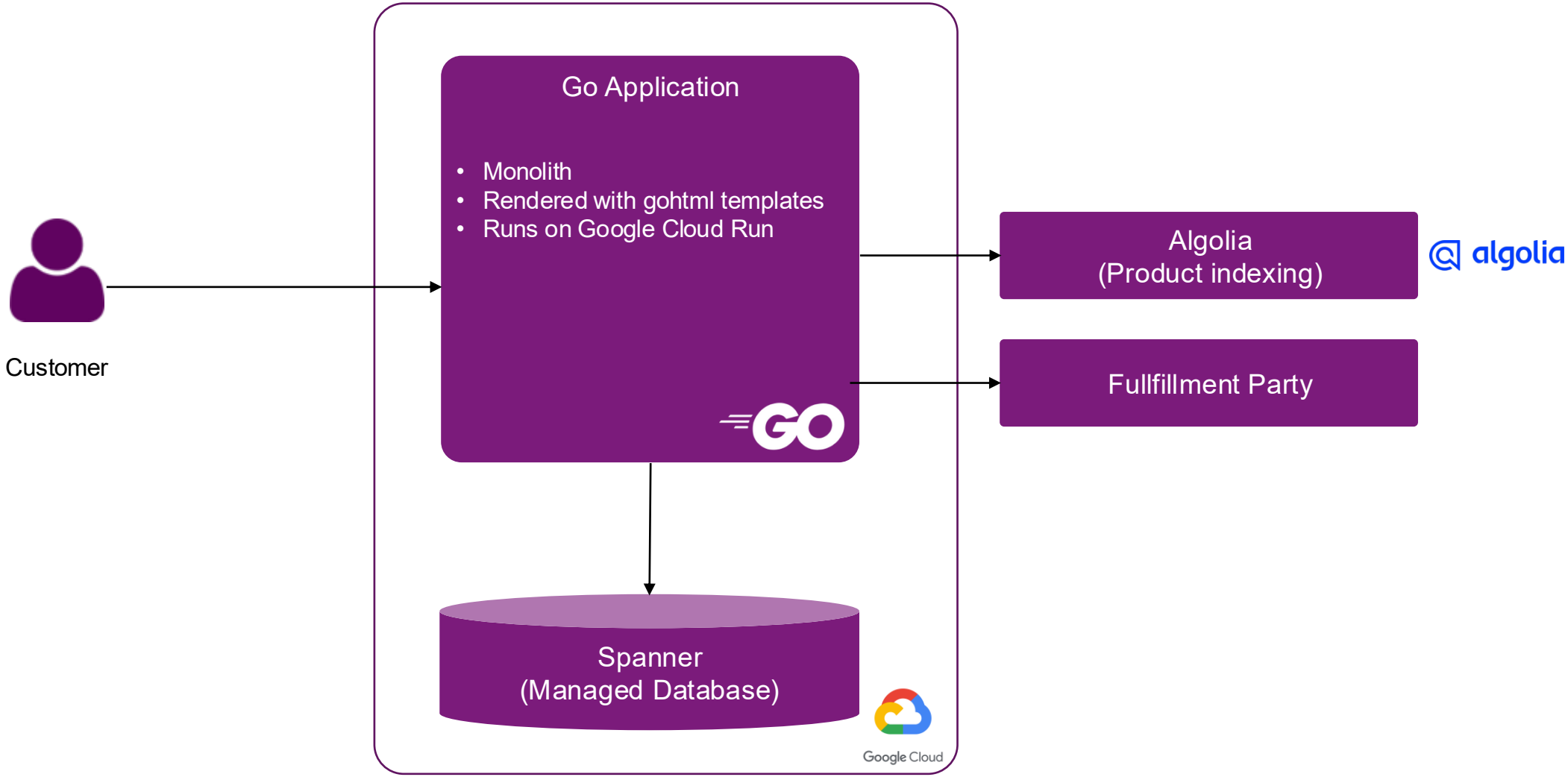
The project that changed everything



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What I actually saw:

- visible system shape
- low operational drag
- code that was easy to inspect
- a team comfortable owning the whole product

Go was not the whole reason the team stayed lean,
but it made the lean path easier to choose and harder to accidentally abandon.

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2 3 Things Go Does Differently

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#1 Go makes it harder to hide complexity

- Application wiring stays visible
- Dependency construction stays visible
- Configuration stays visible
- Startup behaviour stays visible

Framework-driven construction buys convenience. (Java/Kotlin + Spring)

Plain-code construction buys inspectability. (Vanilla go)

On this team, inspectability was worth more.

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#1 Go makes it harder to hide complexity

Kotlin + Spring Application Wiring

Application.kt

```kotlin
@SpringBootApplication
class App

fun main(args: Array<String>) {
    runApplication<App>(*args)
}
```

PaymentConfig.kt

```kotlin
@Configuration
class PaymentConfig {
    @Bean
    @Profile("prod")
    fun paymentClientProd(
        @Value("\\${payment.base-url}") baseUrl: String,
        @Value("\\${payment.api-key}") apiKey: String
    ): PaymentClient = RealPaymentClient(baseUrl, apiKey)

    @Bean
    @Profile("dev")
    fun paymentClientStub(): PaymentClient = StubPaymentClient()
}
```

OrderService.kt

```kotlin
@Service
class OrderService(
    private val paymentClient: PaymentClient,
    private val orderRepo: OrderRepository
) {
    fun submit(orderId: String) {
        val order = orderRepo.find(orderId)
        paymentClient.charge(order.total)
    }
}
```

OrderController.kt

```kotlin
@RestController
class OrderController(
    private val orderService: OrderService
) {
    @PostMapping("/orders/{id}/submit")
    fun submit(@PathVariable id: String) {
        orderService.submit(id)
    }
}
```

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#1 Go makes it harder to hide complexity

Go Application Wiring

main.go

```go
type Config struct {
    Env, HTTPAddr, PaymentBaseURL, PaymentAPIKey string
}
func mustLoadConfig() Config { /* reads env vars with fallbacks — all explicit */ }

func NewOrderService(payments PaymentClient, repo *OrderRepository, ...) *OrderService

func main() {
    cfg := mustLoadConfig()
    var paymentClient PaymentClient
    if cfg.Env == "prod" {
        paymentClient = NewRealPaymentClient(cfg.PaymentBaseURL, cfg.PaymentAPIKey)
    } else {
        paymentClient = NewStubPaymentClient()
    }
    orderSvc := NewOrderService(paymentClient, NewOrderRepository(), logger)
    r.Post("/orders/{id}/submit", SubmitOrderHandler(orderSvc))
    http.ListenAndServe(cfg.HTTPAddr, r)
}
```

service.go

```go
type PaymentClient interface { Charge(ctx context.Context, amount int64) error }

func NewOrderService(payments PaymentClient, repo *OrderRepository) *OrderService {
  return &OrderService{ payments: payments, repo: repo}
}

func (s *OrderService) Submit(ctx context.Context, orderID string) error {
  order,err  := s.repo.find(orderID)
  if err != nil { return fmt.Errorf("finding order: %w", err) }
  return s.payments.Charge(ctx, orderID)
}
```

routes.go

```go
func SubmitOrderHandler(svc *OrderService) http.HandlerFunc {
  return func(w http.ResponseWriter, r *http.Request) {
    if err := svc.Submit(r.Context(), chi.URLParam(r, "id")); err != nil {
      http.Error(w, "submit failed", http.StatusInternalServerError)
      return
    }
    w.WriteHeader(http.StatusNoContent)
  }
}
```

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#2 Explicit error and transaction boundaries force decisions into the open

- Error handling is explicit
- Retries do not happen by magic
- Fallbacks do not happen by magic
- Transaction scope does not happen by magic

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#2 Explicit error and transaction boundaries force decisions into the open

Product page can degrade. Checkout cannot.

product.go

```go
price, err := h.prices.LivePrice(r.Context(), id)
if err != nil {
    h.logger.Warn("...", err)

    price, err = h.cache.CachedPrice(r, id)
    if err != nil {
        http.Error(w, "...", http.StatusServiceUnavailable)
        return
    }
}
// Stale price may be acceptable here.
```

checkout.go

```go
price, err := h.prices.LivePrice(r.Context(), id)
if err != nil {
    h.logger.Error("...", err)

    http.Error(w, "...", http.StatusConflict)
    return
}


// Same failure. No fallback here.
// Policy is visible, local, explicit.
```

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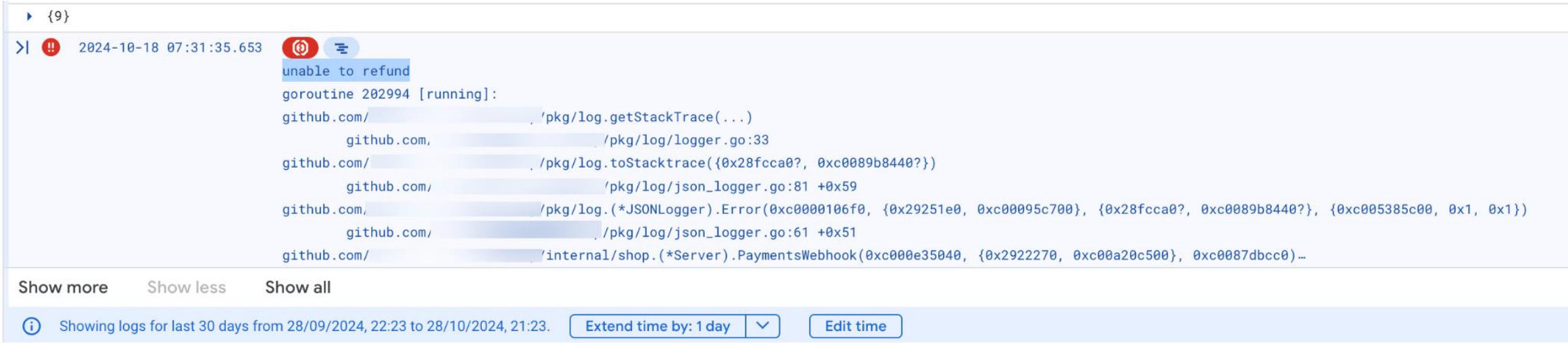
Production incident: a slow query turned retries into duplicate refunds



Some refund payments requests has been rejected by Adyen

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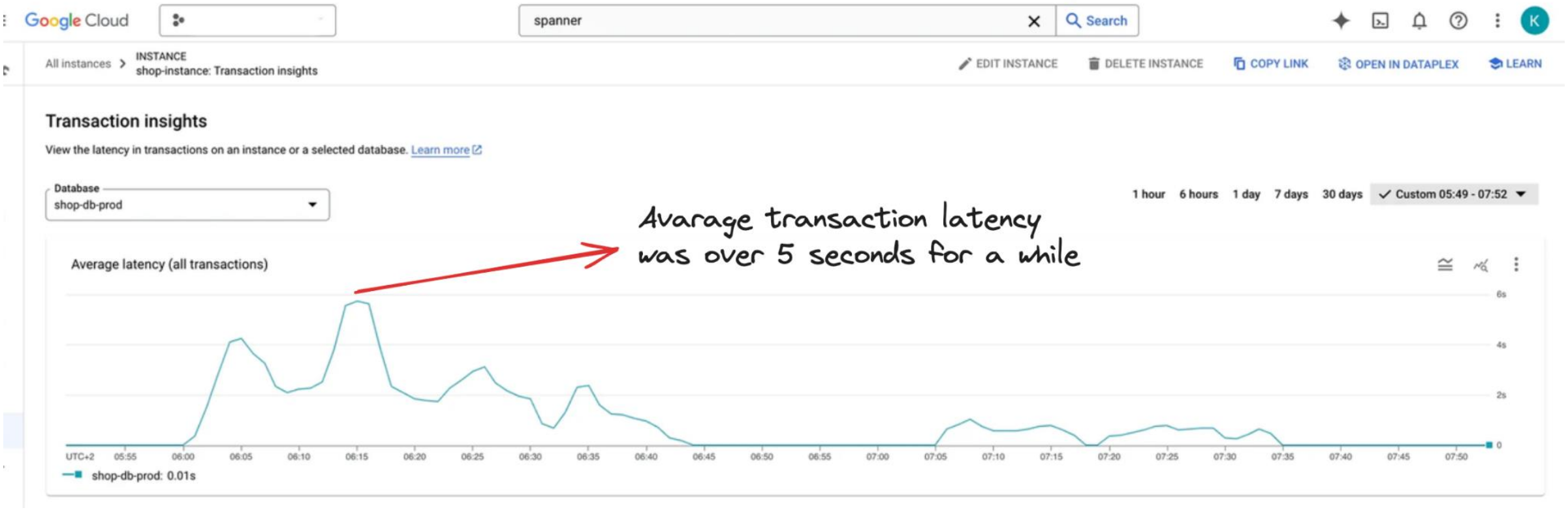
Production incident: a transaction bug locked the database



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Production incident: a transaction bug locked the database

db.go

```go
func main(){
  ...
  messaging.Subscribe(ctx, "refund_payments", func(ctx context.Context, msg RefundEvent){
    refund, err := db.Query("SELECT * FROM refunds WHERE refund_id = ?", msg.RefundID)
    if err != nil { return err }

    if refund.PaymentReference != "" { return nil }

    paymentRef, err := adyenRefund(ctx, refund)
    if err != nil { return err }

    err = db.ReadWriteTransaction(ctx, func(ctx context.Context, tx *sql.Tx) error {
      _ , err := tx.Exec("UPDATE refunds SET paymentRef = ? WHERE refund_id = ?",
                 paymentRef, msg.RefundID)
      return err
    })

    if err != nil { return err }
    return nil
  })
  ...
}
```

Bug #1 Inefficient query caused table-wide locks
refunds table became slow and blocked writes

Bug #2 Missing idempotency key
retries re-issued the refund
PSP treated each retry as a new request

Result: refund succeeded but was not recorded
refund payments happen but not marked as done

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Production incident: a transaction bug locked the database

db.go

```go
func main(){
  ...
  messaging.Subscribe(ctx, "refund_payments", func(ctx context.Context, msg RefundEvent){
    refund, err := db.Query("SELECT * FROM refunds WHERE refund_id = ?", msg.RefundID)
    if err != nil { return err }

    if refund.PaymentReference != "" { return nil }

    paymentRef, err := adyenRefund(ctx, refund)
    if err != nil { return err }

    err = db.ReadWriteTransaction(ctx, func(ctx context.Context, tx *sql.Tx) error {
      _ , err := tx.Exec("UPDATE refunds SET paymentRef = ? WHERE refund_id = ?",
                 paymentRef, msg.RefundID)
      return err
    })

    if err != nil { return err }
    return nil
  })
  ...
}
```

Event handler scope
Retries entire operation on any error

Database read transaction scope

Read-write transaction scope
Retries database work on error

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Production incident: a transaction bug locked the database

In Java + Spring

- Transaction scope often starts at @Transactional
- Framework and proxy behavior matter
- Reconstruction cost is higher

In Go

- Transaction scope usually starts where code says it starts
- Control flow is local
- Debugging path is shorter

The bug was ours. What Go improved was the debugging surface.

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#3 Dependency culture reduces maintenance drag

- Smaller dependency surface
- Less dependent on any framework
- Boring updates

A lean team does not just need fast feature delivery.

It also needs upgrades and maintenance to be boring.

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#3 Dependency culture reduces maintenance drag

In the Go project:

- a small set of major dependencies
- external client libraries for payment/search/fulfillment
- router
- database client
- Most updates uneventful

In JVM Projects

- Major version updates are painful
- Breaking changes are plenty
- Common to see projects stuck in very old version stacks
- Framework versions adds another level complexity

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3 What I Still Miss

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What I still miss from richer languages

JSON APIs: omitted vs null vs zero value

```go
type UpdateUserRequest struct {
    DisplayName    string `json:"displayName"`
    Phone          string `json:"phone"`
    MarketingOptIn bool   `json:"marketingOptIn"`
}

// Was marketingOptIn omitted, or set to false?
// Was displayName omitted, or intentionally cleared?
```

```go
type UpdateUserRequest struct {
    DisplayName    *string `json:"displayName"`
    Phone          *string `json:"phone"`
    MarketingOptIn *bool   `json:"marketingOptIn"`
}

// Distinguishable — but now everything
// is a pointer. Nil checks everywhere.
```

- Transport structs get noisy
- Validation gets noisier
- Business code gets nil checks throughout

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What I still miss from richer languages

Rich domain modeling: where sealed hierarchies help

PaymentResult.kt

```kotlin
sealed interface PaymentResult {
    data class Authorized(
        val authId: String,
        val expiresAt: Instant) : PaymentResult
    data class Declined(val reason: String)
    data class Requires3DS(val redirectUrl: String)
    data class Failed(val errorCode: String)
}
fun handle(result: PaymentResult) = when (result) {
    is Authorized   -> capture(result.authId)
    is Declined     -> showDecline(result.reason)
    is Requires3DS  -> redirect(result.redirectUrl)
    is Failed       -> logFailure(result.errorCode)
}  // Exhaustive — compiler enforced.
```

payment_result.go

```go
type PaymentStatus string
const (
    PaymentAuthorized  PaymentStatus = "authorized"
    PaymentDeclined    PaymentStatus = "declined"
    PaymentRequires3DS PaymentStatus = "requires_3ds"
    PaymentFailed      PaymentStatus = "failed"
)
type PaymentResult struct {
    Status      PaymentStatus
    AuthID      string    // only valid when Authorized
    Reason      string    // only valid when Declined
    RedirectURL string    // only valid when Requires3DS
    ErrorCode   string    // only valid when Failed
}
// Invalid combinations are easy to construct.
```

In Go, the usual answer is disciplined constructors, validation, and tests.
That works. It is just less help from the compiler.

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4 What Actually Changed My Mind

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What actually changed my mind

Lesson 1: Asking the right questions

I used to ask:

Which language lets me express more?

Now I **also** ask

Which language helps a team safely understand, change and run the whole system?

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What actually changed my mind

Lesson 1: Asking the right questions

Lesson 2: There is no best language but tradeoffs

- Best language and tools depends on the team, project and many other factors
- Often there could be more than one good solution for same circumstances

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What actually changed my mind

Lesson 1: Asking the right questions

Lesson 2: There is no best language but tradeoffs

Lesson 3: Adapting a new language extends your views on other languages

- Working with Go made me realise design choices and tradeoffs in Kotlin and Java better

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What actually changed my mind

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What changed my mind as a Java/Kotlin developer?

Lesson 1: Asking the right questions

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